Crash-consistent Snapshot copies

Snapdrive for Unix

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Crash-consistent Snapshot copies

You might have to create a crash-consistent Snapshot copies of your file system or disk groups. SnapDrive for UNIX creates Snapshot copies that contain the image of all the storage system volumes specified in the entity.

When you create a Snapshot copy of a storage entity, such as a file system or disk group, SnapDrive for UNIX creates a Snapshot copy that contains the image of all the storage system volumes that comprise the entity you specified using a `file_spec` argument. The `file_spec` argument specifies the storage entity, such as the file system, LUN, or NFS directory tree that SnapDrive for UNIX uses to create the Snapshot copy.

SnapDrive for UNIX makes consistent storage components that compose the entity you requested in the Snapshot copy. This means that LUNs or directories being used outside those specified by the `snapdrive snap create` command `file_spec` argument might not have consistent images in the Snapshot copy. SnapDrive for UNIX enables you to restore only the entities specified by the `file_spec` argument that are consistent in the Snapshot copy.

Snapshot copies of entities contained on a single storage system volume are always crash-consistent. SnapDrive for UNIX takes special steps to ensure that Snapshot copies that span multiple storage systems or storage system volumes are also crash-consistent. The method that SnapDrive for UNIX uses to ensure crash consistency depends on the Data ONTAP version used where the storage entities in your Snapshot copy resides.

Crash consistency with Data ONTAP 7.2 and later

SnapDrive for UNIX uses the support for consistency groups provided by Data ONTAP 7.2 and later versions, such that all Snapshot copies that span multiple volumes are crash consistent.

Data ONTAP versions 7.2 and greater provides support for consistency groups and storage system fencing. SnapDrive for UNIX uses these features to ensure that all Snapshot copies that span multiple volumes are crash consistent.

To create a crash consistent Snapshot copy across multiple volumes, SnapDrive for UNIX does the following:

- Fences (freezes) I/O to every volume that contains a storage entity.
- Takes a Snapshot copy of each volume.

The time it takes to fence the volume and create the Snapshot copy is limited, and is controlled by Data ONTAP.

The **`snapcreate-cg-timeout`** parameter in the `snapdrive.conf` file specifies the amount of time, within Data ONTAP limitations, that you wish to allow for storage system fencing. You can specify an interval that is urgent, medium, or relaxed. If the storage system requires more time than allowed to complete the fencing operation, SnapDrive for UNIX creates the Snapshot copy using the consistency methodology for previous Data ONTAP 7.2 versions. You can also specify this methodology by using the `-nofilerfence` option when you create the Snapshot copy.

If you request a Snapshot copy for a storage entity that spans storage systems with both Data ONTAP 7.2 and previous Data ONTAP versions, SnapDrive for UNIX also creates the Snapshot copy using the consistency method for Data ONTAP versions before 7.2.

Consistency group Snapshot copies in SnapDrive for UNIX

Consistency Group Snapshot is a Snapshot copy of a set of volumes that span different Vservers or nodes of a cluster, which is managed as a single entity.

SnapDrive for UNIX captures crash-consistent Snapshot copies across all volumes spanning different Vservers or nodes of a cluster. You can also configure the time period within which the Snapshot copy is to be captured.

SnapDrive for UNIX captures consistency group Snapshot copies by default. You can disable this feature and revert to capturing Snapshot copies in best-effort mode.

> SnapDrive for UNIX 5.2 supports consistency group Snapshot copies for clustered Data ONTAP only in Data ONTAP 8.2 or later versions.

Related information

Creating a consistency group Snapshot copy

Disabling consistency group Snapshots copies

Creating a consistency group Snapshot copy

You can configure SnapDrive for UNIX to create a consistency group Snapshot copy.

Steps

1. Enter the following command on the host:

   `snapdrive snap create -fs /mnt/test -snapname` *`snapshotname`* `-f -noprompt`.

   `snapshotname` is the name specified for the consistency group Snapshot copy.

Example

The following is an example of the command:

```
snapdrive snap create -fs /mnt/test -snapname snap_123 -f -noprompt
```

The consistency group Snapshot copy is successfully created.

Disabling consistency group Snapshots copies

You can configure SnapDrive for UNIX to disable a consistency group Snapshot copy.

Steps

1. Enter:

   `snapdrive snap create -fs /mnt/fs3 -`*`snapname`*` nfs_snap -`*`nofilerfence`*

   The consistency group Snapshot copy is successfully disabled.

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